

Experiment 15: Decision Tree

Aim

To implement a Decision Tree classifier using Python and classify the Iris dataset.

Algorithm

1. Import the required libraries.
 2. Load the Iris dataset.
 3. Split the dataset into training and testing sets.
 4. Create a Decision Tree classifier.
 5. Train the model using the training data.
 6. Predict the class labels for the test data.
 7. Calculate and display the accuracy of the model.
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Python Code

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score

# Load dataset
iris = load_iris()
X = iris.data
y = iris.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
```

```
# Create Decision Tree model
model = DecisionTreeClassifier()

# Train the model
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Display results
print("Predicted Values:", y_pred)
print("Actual Values:", y_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
```

output:

```
... Predicted Values: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
  0 0 0 2 1 1 0 0]
Actual Values      : [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
  0 0 0 2 1 1 0 0]
Accuracy: 1.0
```

Result

The Decision Tree classifier was successfully implemented using Python. The model classified the Iris dataset with high accuracy (approximately **95%–100%**, depending on the train-test split).

Experiment 16: Feed Forward Neural Network

Aim

To implement a Feed Forward Neural Network (FFNN) using Python for classifying the Iris dataset.

Algorithm

1. Import the required libraries.
 2. Load the Iris dataset.
 3. Convert the output labels into one-hot encoding.
 4. Split the dataset into training and testing sets.
 5. Create a Feed Forward Neural Network with one hidden layer.
 6. Compile the model using the Adam optimizer.
 7. Train the model using the training dataset.
 8. Evaluate the model and display the classification accuracy.
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Python Code

```
import tensorflow as tf

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
from tensorflow.keras.utils import to_categorical
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split

# Load dataset
iris = load_iris()
X = iris.data
y = iris.target

# One-hot encoding
```

```
y = to_categorical(y)
```

```
# Split dataset
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.3, random_state=42  
)
```

```
# Build Feed Forward Neural Network
```

```
model = Sequential()
```

```
model.add(Dense(10, activation='relu', input_shape=(4,)))
```

```
model.add(Dense(3, activation='softmax'))
```

```
# Compile model
```

```
model.compile(  
    optimizer='adam',  
    loss='categorical_crossentropy',  
    metrics=['accuracy']  
)
```

```
# Train model
```

```
model.fit(X_train, y_train, epochs=50, batch_size=5, verbose=0)
```

```
# Evaluate model
```

```
loss, accuracy = model.evaluate(X_test, y_test, verbose=0)
```

```
print("Accuracy:", accuracy)
```

output:

... Accuracy: 0.977777791023254

Result

The Feed Forward Neural Network was successfully implemented using Python. The model learned the patterns in the Iris dataset and achieved a classification accuracy of approximately **95%–99%** on the test data.